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Corner module apparatus for vehicle

Abstract

A corner module apparatus for a vehicle including a driving unit installed inside a wheel and configured to provide a driving force to the wheel, a first knuckle coupled to the driving unit, a second knuckle spaced apart from the first knuckle in a width direction of the vehicle and disposed to face each other, a suspension unit configured to connect to the second knuckle and supports the second knuckle for a vehicle body, a steering driving unit installed in the second knuckle configured to generate a steering force, and a steering angle adjustment unit connected to the first knuckle and configured to adjust a steering angle of the wheel in conjunction with the steering force generated from the steering driving unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the benefit under 35 USC § 119 (a) of Korean Patent Application No. 10-2022-0014080, filed on Feb. 3, 2022, and Application No. 10-2022-0060770, filed on May 18, 2022, the entire disclosures of which are hereby incorporated by reference for all purposes as if set forth herein.

FIELD

(2) Exemplary embodiments of the present disclosure relate to a corner module apparatus for a vehicle, and more particularly, in which a driving, a braking, a steering, and a suspension system are integrated.

DESCRIPTION OF THE RELATED ART

(3) In general, an electric vehicle is an eco-friendly vehicle that does not emit exhaust gas, and is equipped with a high voltage battery that supplies energy for driving, a driving motor that generates

rotational force from power output from the high voltage battery, and a rotational power of a motor is transmitted to a wheel through a drive shaft.

(4) In recent years, in-wheel motor vehicles have been in the spotlight, which reduce weight of a vehicle by omitting intermediate power transmission devices such as decelerators and differential gears, and consider an advantage of reducing energy loss in a power transmission process, and power of a motor is transmitted directly to a wheel because the motor is directly installed inside the wheel. Furthermore, not only a driving system but also a braking, a steering, and a suspension system are being actively developed.

SUMMARY

(5) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(6) In one general aspect, here is provided a corner module apparatus for a vehicle including a driving unit installed inside a wheel, the driving unit being configured to provide a driving force to the wheel, a first knuckle coupled to the driving unit, a second knuckle spaced apart from the first knuckle in a width direction of the vehicle and disposed to face each other, a suspension unit configured to connect to the second knuckle and to support the second knuckle for a vehicle body, a steering driving unit installed in the second knuckle configured to generate a steering force, and a steering angle adjustment unit connected to the first knuckle, the steering angle adjustment unit being configured to adjust a steering angle of the wheel in conjunction with the steering force generated from the steering driving unit.

(7) The second knuckle may include a knuckle body disposed to face the first knuckle and provided with an accommodation unit into which the steering driving unit is inserted, a mounting unit which extends from a first side of the knuckle body, the mounting unit being configured to support the steering angle adjustment unit, a first connection unit extending from the mounting unit and connected to the suspension unit, and a second connection unit extending from a second side of the knuckle body and connected to the suspension unit.

(8) The knuckle body may be disposed to be inclined with respect to a ground.

(9) The accommodation unit may penetrate the knuckle body in a width direction of the vehicle.

(10) The first connection unit and the second connection unit may be disposed to be spaced apart from each other in a direction perpendicular to a ground.

(11) The steering angle adjustment unit may include a deceleration unit configured to connect with the second knuckle and to slow down the steering force generated from the steering driving unit and a joint unit configured to vary the steering angle of the wheel by transmitting the steering force from the deceleration unit to the first knuckle.

(12) The deceleration unit may include a first transmission gear configured to rotate with an input shaft of the steering driving unit, a second transmission gear that is configured to engage with the first transmission gear and to rotate in conjunction with a rotation of the first transmission gear, and a third transmission gear configured to engage with the second transmission gear and to rotate an output shaft in conjunction with a rotation of the second transmission gear.

(13) The first transmission gear may include a shape of a worm shaft and worm threads provided on an outer peripheral surface of the first transmission gear.

(14) The joint unit includes a first joint extending from the output shaft and connected to a first side of the first knuckle and a second joint extending from the second knuckle, spaced apart from the first joint, and connected to a second side of the first knuckle.

(15) The first joint and the second joint may be disposed to be inclined at a predetermined angle with respect to a ground.

(16) The first joint and the second joint may be constant velocity joints.

(17) The suspension unit may include a suspension arm provided between the second knuckle and

the vehicle body and being configured to support the second knuckle and a shock absorber connected to the suspension arm, the shock absorber configured to absorb an impact transmitted from a road surface.

(18) The suspension arm may include a first arm may include a first end rotatably connected to the first connection unit and a second end rotatably connected to the vehicle body and a second arm, the second arm being spaced apart from the first arm, the second arm may include a third end rotatably connected to the second connection unit and a fourth rotatably connected to the vehicle body.

(19) In another general aspect, here is provided a corner module apparatus for a vehicle including a driving unit installed inside a wheel, the driving unit being configured to provide a driving force to the wheel, a first knuckle coupled to the driving unit, a second knuckle spaced away from the first knuckle to face the first knuckle in a width direction of the vehicle and being configured to rotatably support the first knuckle, a suspension arm configured to connect the second knuckle and a vehicle body and configured to support the second knuckle with respect to the vehicle body, a joint arm configured to connect the first knuckle and the second knuckle to each other, and an angle adjustment unit provided between the second knuckle and the joint arm and configured to adjust a relative angle between the first knuckle and the second knuckle by axially rotating the joint arm in the width direction of the vehicle.

(20) The joint arm may include a ball joint rotatably jointed to the first knuckle, a fastener fastened to the second knuckle, and a connection member configured to connect, the connection member being provided between the ball joint and the fastener.

(21) A first fastening hole may be defined in the joint arm, the first fastening hole passing through an outer surface of the fastener and a first slot hole may also be defined in the joint arm, the first slot hole being positioned under the first fastening hole and passing through the outer surface of the fastener.

(22) A second fastening hole may be defined in the second knuckle, the second fastening hole corresponding to the first fastening hole and a second slot hole may be defined in the second knuckle, the second slot hole corresponding to the first slot hole.

(23) The first slot hole and the second slot hole may be formed in a shape of a longitudinal hole in the width direction of the vehicle.

(24) The angle adjustment unit may include a cam bolt, a washer being eccentrically coupled to an outer circumferential surface of a bolt shaft of the cam bolt, wherein the first slot hole and the second slot hole pass through the bolt shaft and a guide unit formed on an outer surface of the second knuckle, the guide unit being configured to guide the washer in such a manner as to eccentrically rotate the cam bolt.

(25) The angle adjustment unit may also include an adjustment member passing through the first fastening hole and the second fastening hole, the adjustment member being configured to fasten the second knuckle and the fastener to each other.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view schematically illustrating a configuration of a corner module apparatus for a vehicle according to a first embodiment of the present disclosure.

(2) FIG. 2 is a side view schematically illustrating the configuration of the corner module apparatus for a vehicle according to the first embodiment of the present disclosure.

(3) FIG. 3 is an exploded perspective view schematically illustrating the configuration of the corner module apparatus for a vehicle according to the first embodiment of the present disclosure.

(4) FIG. 4 is a perspective view schematically illustrating a configuration of a second knuckle unit

according to the first embodiment of the present disclosure.

(5) FIG. 5 is a perspective view schematically illustrating a configuration of a steering driving unit and a steering angle adjustment unit according to the first embodiment of the present disclosure.

(6) FIG. 6 is a cross-sectional view schematically illustrating the configuration of the steering drive unit and the steering angle adjustment unit according to the first embodiment of the present disclosure.

(7) FIG. 7 is an enlarged view schematically illustrating a configuration of a first joint according to the first embodiment of the present disclosure.

(8) FIG. 8 is an enlarged view schematically illustrating a configuration of a second joint according to the first embodiment of the present disclosure.

(9) FIG. 9 is a perspective view schematically illustrating a configuration of a corner module apparatus for a vehicle according to a second embodiment of the present disclosure.

(10) FIG. 10 is a front view schematically illustrating a configuration of the corner module apparatus for a vehicle according to the second embodiment of the present disclosure.

(11) FIG. 11 is an exploded perspective view schematically illustrating a configuration of the corner module apparatus for a vehicle according to the second embodiment of the present disclosure.

(12) FIG. 12 is an exploded perspective view illustrating an angle adjustment unit in the corner module apparatus for a vehicle according to the second embodiment of the present disclosure.

(13) FIG. 13 is a view illustrating a state where the corner module apparatus for a vehicle according to the second embodiment of the present disclosure is operated for zero camber setting.

(14) FIG. 14 is a view illustrating a state where the corner module apparatus for a vehicle according to the second embodiment of the present disclosure is operated for negative camber setting.

(15) FIG. 15 is a view illustrating a state where the corner module apparatus for a vehicle according to the second embodiment of the present disclosure is operated for positive camber setting.

(16) Throughout the drawings and the detailed description, unless otherwise described or provided, the same drawing reference numerals will be understood to refer to the same elements, features, and structures. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

(17) The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent after an understanding of the disclosure of this application. For example, the sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed as will be apparent after an understanding of the disclosure of this application, with the exception of operations necessarily occurring in a certain order.

(18) The features described herein may be embodied in different forms and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided merely to illustrate some of the many possible ways of implementing the methods, apparatuses, and/or systems described herein that will be apparent after an understanding of the disclosure of this application.

(19) Advantages and features of the present disclosure and methods of achieving the advantages and features will be clear with reference to embodiments described in detail below together with the accompanying drawings. However, the present disclosure is not limited to the embodiments disclosed herein but will be implemented in various forms. The embodiments of the present disclosure are provided so that the present disclosure is completely disclosed, and a person with ordinary skill in the art can fully understand the scope of the present disclosure. The present disclosure will be defined only by the scope of the appended claims. Meanwhile, the terms used in the present specification are for explaining the embodiments, not for limiting the present

disclosure.

(20) Terms, such as first, second, A, B, (a), (b) or the like, may be used herein to describe components. Each of these terminologies is not used to define an essence, order or sequence of a corresponding component but used merely to distinguish the corresponding component from other component(s). For example, a first component may be referred to as a second component, and similarly the second component may also be referred to as the first component.

(21) Throughout the specification, when a component is described as being “connected to,” or “coupled to” another component, it may be directly “connected to,” or “coupled to” the other component, or there may be one or more other components intervening therebetween. In contrast, when an element is described as being “directly connected to,” or “directly coupled to” another element, there can be no other elements intervening therebetween.

(22) In a description of the embodiment, in a case in which any one element is described as being formed on or under another element, such a description includes both a case in which the two elements are formed in direct contact with each other and a case in which the two elements are in indirect contact with each other with one or more other elements interposed between the two elements. In addition, when one element is described as being formed on or under another element, such a description may include a case in which the one element is formed at an upper side or a lower side with respect to another element.

(23) The singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises/comprising” and/or “includes/including” when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components and/or groups thereof.

(24) FIG. 1 is a perspective view schematically illustrating a configuration of a corner module apparatus for a vehicle according to first embodiment of the present disclosure, FIG. 2 is a side view schematically illustrating the configuration of the corner module apparatus for a vehicle according to first embodiment of the present disclosure, and FIG. 3 is an exploded perspective view schematically illustrating the configuration of the corner module apparatus for a vehicle according to first embodiment of the present disclosure.

(25) Referring to FIGS. 1 and 2, a corner module apparatus for a vehicle 1 according to the present disclosure includes a driving unit 100, a braking unit 200, a first knuckle 310, the second knuckle 320, a suspension unit 400, a steering driving unit 500, and a steering angle adjustment unit 600.

(26) The driving unit 100 is installed inside a wheel 2 of a vehicle and rotates the wheel 2 by providing a driving force to the wheel 2. The driving unit 100 is installed on each of the wheels 2 of the vehicle to individually provide driving force to a plurality of wheels 2. The driving unit 100 according to the present disclosure may include a stator fixed inside the wheel 2, which forms a magnetic field by receiving power from a vehicle battery, and a rotor installed to be rotated inside the wheel 2, and rotating the wheel 2 by electromagnetic interaction with the stator. A central axis of the stator and the rotor may be positioned on the same line as a central axis of the wheel 2, and may be disposed to be concentrically stacked inside the wheel 2.

(27) The braking unit 200 is installed inside the wheel 2 and interferes with a rotation of the wheel 2 to apply or release the braking force.

(28) The braking unit 200 according to the present disclosure includes a brake disk 210 and a brake caliper 220.

(29) The brake disk 210 is connected to the wheel 2 or the driving unit 100 and rotates in conjunction with rotation of the wheel 2. The brake disk 210 according to the present disclosure is formed to have the shape of a disk and is installed inside the wheel 2. The brake disk 210 is disposed such that a central axis thereof is aligned with the central axis of the wheel 2. The brake disk 210 may be integrally connected to the rotor of the driving unit 100 or the wheel 2 by bolting

or the like. Accordingly, the brake disk **210** may be rotated about the central axis together with the wheel **2** when the wheel **2** is rotated. A diameter of the brake disk **210** may be variously changed in design according to a diameter of the wheel **2** and a size of the driving unit **100**.

(30) The brake caliper **220** applies a braking force by pressing the brake disk **210** when braking the vehicle. The brake caliper **220** according to of the present disclosure may include a brake pad disposed and facing the brake disk **210**, a caliper housing coupled to the first knuckle **310** to be described later, a caliper housing for supporting the brake pad movably, and a piston which is installed to move forward and backward in the caliper housing and presses or releases the brake pad toward the brake disk **210** according to the moving direction.

(31) The first knuckle **310** is combined with the driving unit **100** and functions as a configuration to form a kingpin shaft, which is a central axis of steering, by providing a mechanical connection to a wheel **2** of a steering angle adjustment unit **600** described later. The first knuckle **310** according to the present disclosure may be bonded and supported by a stator of the driving unit **100** by bolting or the like. The first knuckle **310** may rotatably support the rotor of the driving unit **100** through a wheel bearing or the like. The first knuckle **310** may be manufactured by molding a metal-based material with a casting or the like to ensure sufficient rigidity. The specific shape of the first knuckle **310** is not limited to the shape illustrated in FIGS. **1** and **2**, and is combined with the driving unit **100** to make various changes in design within a technical idea of a shape that may be disposed to face an inner surface of the wheel **2**.

(32) A second knuckle **320** is spaced apart from the first knuckle **310** and supports the steering driving unit **500** and the steering angle adjustment unit **600**, which will be described later, while providing a mechanical connection to a body of a suspension unit **400**. For this reason, the second knuckle **320** functions as a configuration for forming a suspension shaft that moves up and down when the wheel **2** bumps and rebounds. The second knuckle **320** is disposed to face the first knuckle **310** at a predetermined interval in a width direction of the vehicle. Accordingly, the first knuckle **310** and the second knuckle **320** dispose the kingpin shaft to a position close to the wheel **2** by separating the suspension shaft formed in the second knuckle **320** and the kingpin shaft formed in the first knuckle **310** each other in the width direction of the vehicle, thereby reducing an offset value of the kingpin and improving the driving and braking stability of the vehicle.

(33) FIG. **4** is a perspective view schematically illustrating a configuration of a second knuckle according to first embodiment of the present disclosure.

(34) Referring to FIG. **4**, the second knuckle **320** according to the present disclosure includes a knuckle body **321**, a mounting unit **322**, a first connection unit **323**, and a second connection unit **324**.

(35) The knuckle body **321** forms a central appearance of the second knuckle **320** and entirely supports the mounting unit **322**, the first connection unit **323**, and the second connection unit **324**. The knuckle body **321** according to the present disclosure may be provided with a hole-shaped accommodation unit **321a** penetrating the knuckle body **321** in the width direction of the vehicle. Accordingly, the knuckle body **321** may be formed to have substantially the shape of a hollow square frame. The steering driving unit **500** is inserted into the accommodation unit **321a**, and the knuckle body **321** is combined with the steering driving unit **500** inserted into the accommodation unit **321a** by welding and bolting to support the steering driving unit **500**. The knuckle body **321** may be disposed to be inclined at a predetermined angle with respect to the ground such that an output shaft **614** of the steering angle adjustment unit **600** is disposed parallel to the kingpin shaft.

(36) The mounting unit **322** extends from one side of the knuckle body **321** to form an appearance of one side of the second knuckle **320**. The mounting unit **322** is combined with the steering angle adjustment unit **600** to support the steering angle adjustment unit **600**. The mounting unit **322** according to the present disclosure extends from an upper end of the knuckle body **321** toward the wheel **2**. A lower side surface of the mounting unit **322** is formed so that an upper side surface of a deceleration unit **610** in the steering angle adjustment unit **600** may be seated. In a state in which

the upper side surface of the deceleration unit **610** is seated on the lower side surface of the mounting unit **322**, as the mounting unit **322** is integrally coupled with the deceleration unit **610**, the mounting unit **322** supports the steering angle adjustment unit **600** by bolting coupling or the like.

(37) The first connection unit **323** extends from the mounting unit **322** and is connected to a first arm **411** provided in a suspension unit **400**. The first connection unit **323** according to the present disclosure may be formed to have the shape of a ring protruding upward from an upper side surface of the mounting unit **322**. The first connection unit **323** may be connected to one end of the first arm **411** provided in the suspension unit **400** through a bush, a ball joint, a pin, or the like. The first connection unit **323** rotatably supports one end of the first arm **411** to function as an upper reference point of a suspension shaft that moves up and down when the wheel **2** bumps and rebounds.

(38) The second connection unit **324** extends from the other side of the knuckle body **321** to form the other side appearance of the second knuckle **320**. The second connection unit **324** is connected to a second arm **412** provided in the suspension unit **400**. The second connection unit **324** according to the present disclosure may be formed to have the shape of a pair of bars extending downward from the lower end of the knuckle body **321**. The second connection unit **324** may be connected to one end of the second arm **412** provided in the suspension unit **400** through a bush, a ball joint, a pin, or the like. The second connection unit **324** rotatably supports one end of the second arm **412** to function as a lower reference point of the suspension shaft that moves up and down when the wheel **2** bumps and rebounds.

(39) The first connection unit **323** and the second connection unit **324** are disposed to be spaced apart from each other in a direction perpendicular to the ground. Accordingly, the first connection unit **323** and the second connection unit **324** may arrange the direction of the suspension shaft in parallel with the direction of movement of the bump and rebound of the wheel **2**.

(40) The suspension unit **400** is connected to the second knuckle **320** and supports the second knuckle **320** with respect to the vehicle body. The suspension unit **400** is provided to absorb an impact transmitted from a road surface through the wheel **2** when the vehicle travels. Herein, the vehicle body may be exemplified as a chassis frame such as a sub frame (not illustrated) installed below the vehicle.

(41) The suspension unit **400** according to the present disclosure includes a suspension arm **410** and a shock absorber **420**.

(42) The suspension arm **410** is provided between the second knuckle **320** and a vehicle body to support the second knuckle **320**. More specifically, the suspension arm **410** connects the wheel **2** to the vehicle body through the second knuckle **320**, absorbs a load applied from the wheel **2** during driving the vehicle by its own rigidity, and controls the movement of the wheel **2**.

(43) The suspension arm **410** according to the present disclosure may include a first arm **411** and a second arm **412**.

(44) The first arm **411** and the second arm **412** are disposed to face each other while being spaced apart in the vertical direction. One end of the first arm **411** is rotatably connected to the first connection unit **323** and the other end thereof is rotatably connected to the vehicle body. The second arm **412** is rotatably connected to the second connection unit **324**, and the other end thereof is rotatably connected to the vehicle body. In this case, both ends of the first arm **411** and the second arm **412** may be rotatably supported on the first connection unit **323** and the vehicle body, respectively, or the second connection unit **324** and the vehicle body, respectively, through a bush, a ball joint, a pin, or the like. The first arm **411** and the second arm **412** may be formed to have the shape of a double wishbone. Accordingly, the first arm **411** and the second arm **412** may be able to set the negative camber of the wheel **2**, thereby improving the cornering performance of the vehicle and enabling low-phase setting to lower a height.

(45) The shock absorber **420** is connected to the suspension arm **410** and is provided extensively

along the longitudinal direction to absorb impact or vibration transmitted from a road surface to the vehicle body through the wheel 2. The shock absorber 420 according to the present disclosure includes a cylinder 421, a rod 422, and an elastic body 423.

(46) The cylinder 421 extends in the vertical direction and is filled with fluid therein. A lower end portion of the cylinder 421 may pass through the first arm 411 and may be rotatably connected to an upper side surface of the second arm 412.

(47) The rod 422 extends in the longitudinal direction of the cylinder 421. The rod 422 is installed to be slidably moved along the longitudinal direction of the cylinder 421 by inserting a lower side thereof into an upper end of the cylinder 421. The other side of the rod 422 is coupled to a wheel mount (not shown) by bolting or the like. The rod 422 moves to be slid along the longitudinal direction of the cylinder 421 when the wheel 2 bumps and rebounds.

(48) The elastic body 423 is disposed to surround outer surfaces of the cylinder 421 and the rod 422, and a length thereof is variable in conjunction with a slide movement of the rod 422. The elastic body 423 according to the present disclosure may be formed to have the shape of a coil spring which is elastic in the longitudinal direction. Both ends of the elastic body 423 may be supported by being coupled to a lower sheet fixed to the cylinder 421 and an upper sheet fixed to the rod 422, respectively. The elastic body 423 is compressed or stretched when the rod 422 slides, accumulates elastic restoring force, and may offset an impact applied from the road surface by an accumulated elastic restoring force.

(49) The steering driving unit 500 is installed in the second knuckle 320 and then generates a steering force.

(50) FIG. 5 is a perspective view schematically illustrating a configuration of a steering driving unit and a steering angle adjustment unit according to first embodiment of the present disclosure, and FIG. 6 is a cross-sectional view schematically illustrating the configuration of the steering drive unit and the steering angle adjustment unit according to first embodiment of the present disclosure.

(51) Referring to FIGS. 1 to 6, the steering driving unit 500 according to the present disclosure may be exemplified as various types of electric motors that generate rotational force from a power source applied from the outside. The steering driving unit 500 is inserted into an accommodation unit 321a and is integrally coupled and supported with the knuckle body 321 by welding or bolting. The steering driving unit 500 may be connected to a battery of a vehicle to receive power from the battery. The steering driving unit 500 is connected to an ECU of the vehicle or the like, and whether a rotational force is generated and a direction of the rotational force or the like may be controlled by a control signal of the ECU.

(52) The steering driving unit 500 is equipped with an input shaft 501 that transmits the rotational force generated by the steering driving unit 500 to a steering angle adjustment unit 600, which will be described later. The input shaft 501 according to the present disclosure is formed to have the shape of an approximate bar and protrudes from an upper end of the steering driving unit 500 toward the mounting unit 322.

(53) The steering angle adjustment unit 600 is connected to the first knuckle 310 and adjusts the steering angle of the wheel 2 in conjunction with a steering force generated by the steering driving unit 500.

(54) The steering angle adjustment unit 600 according to the present disclosure includes a deceleration unit 610 and a joint unit 620.

(55) The deceleration unit 610 is coupled to and supported by the second knuckle 320. More specifically, an upper side of the deceleration unit 610 is seated on a lower side of the mounting unit 322, and the deceleration unit 610 is integrally coupled and supported with the mounting unit 322 by bolting coupling or the like. The deceleration unit 610 decelerates and outputs a steering force generated from the steering driving unit 500. That is, the deceleration unit 610 is provided to amplify a magnitude of the steering force transmitted to the wheel 2 by decelerating a rotation

speed of the input shaft **501** rotated in conjunction with operation of the steering driving unit **500** to a set deceleration ratio.

(56) The deceleration unit **610** according to the present disclosure includes a first transmission gear **611**, a second transmission gear **612**, and a third transmission gear **613**.

(57) The first transmission gear **611** is rotated together with the input shaft **501** of the steering driving unit **500**. The first transmission gear **611** according to the present disclosure may be formed to have the shape of a worm shaft equipped with worm threads on the outer peripheral surface.

Accordingly, when the first transmission gear **611** engages with the second transmission gear **612**, it can be prevented from being reversed by the rotational force transmitted from the second transmission gear **612**, thereby preventing the steering angle of the wheel **2** from being arbitrarily changed. The worm thread formed on the outer peripheral surface of the first transmission gear **611** may vary in a pitch diameter along an axial direction of the first transmission gear **611**. As illustrated in FIG. **6**, the first transmission gear **611** may be disposed parallel to the input shaft **501** as the input shaft **501** is inserted directly through the central axis of the steering driving unit **500**, and may be disposed perpendicular to the input shaft **501** through a separate gear connection structure.

(58) The second transmission gear **612** is engaged with the first transmission gear **611** for being combined therewith and rotated in conjunction with a rotation of the first transmission gear **611**. In the second transmission gear **612** according to the present disclosure, a direction of the central axis is arranged parallel to the input shaft **501**. As illustrated in FIG. **6**, the second transmission gear **612** may be formed to have the shape of a spiral tooth that is conjugated to the worm thread formed in the first transmission gear **611** on the outer peripheral surface. A tooth shape formed on the outer peripheral surface of the second transmission gear **612** may vary in a pitch diameter along an axial direction of the second transmission gear **612**.

(59) In contrast, when the first transmission gear **611** is disposed perpendicular to the input shaft **501**, the second transmission gear **612** may be formed to have a conventional worm wheel shape in which the central axis is disposed parallel to the input shaft **501** and engaged with the worm thread of the first transmission gear **611**.

(60) The third transmission gear **613** is engaged with the second transmission gear **612** for being combined therewith and rotates an output shaft **614** in conjunction with a rotation of the second transmission gear **612**. In the third transmission gear **613** according to the present disclosure, a direction of the central axis is arranged parallel to the input shaft **501**, and the output shaft **614** is inserted directly through the central axis. Accordingly, the central axis of the output shaft **614** is disposed parallel to the input shaft **501** and may be integrally rotated with the third transmission gear **613**. As illustrated in FIG. **6**, the output shaft **614** may be formed on the outer peripheral surface of the third transmission gear **613** to have the shape of a spiral tooth that is conjugated to the tooth shape formed in the second transmission gear **612**. A pitch diameter of the tooth shape formed on the outer peripheral surface of the third transmission gear **613** may vary along the axial direction of the third transmission gear **613**.

(61) In contrast, the third transmission gear **613** may be formed to have a typical helical gear, spur gear, or the like that is engaged with the outer peripheral surface of the second transmission gear **612** when the second transmission gear **612** is formed in the shape of a worm wheel.

(62) A joint unit **620** provides a mechanical connection of the second knuckle **320** and the first knuckle **310**, and at the same time finally transmits the steering force output from the delivery unit **510** to the first knuckle **310** to change the steering angle of the wheel **2**.

(63) The joint unit **620** according to the present disclosure includes a first joint **621** and a second joint **622**.

(64) The first joint **621** extends from the output shaft **614** and is connected to one side of the first knuckle **310**. The first joint **621** finally transmits the steering force generated from the steering driving unit **500** to the first knuckle **310**. In this case, the first joint **621** functions as an upper

reference point of the kingpin shaft, which is a central axis on which the wheel **2** rotates during a steering operation of the wheel **2**.

(65) FIG. **7** is an enlarged view schematically illustrating a configuration of a first joint according to first embodiment of the present disclosure.

(66) Referring to FIGS. **1** to **7**, both sides of the first joint **621** according to the present disclosure are connected to the lower end of the output shaft **614** and the upper side of the first knuckle **310**. The first joint **621** partially allows cutting of the output shaft **614** and the first knuckle **310** according to the vertical behavior of the wheel **2**, at the same time, the first joint **621** may be exemplified as various types of constant speed joints so that the output shaft **614** and the first knuckle **310** may be rotated at the same angular speed.

(67) The second joint **622** extends from the second knuckle **320** and is connected to the other side of the first knuckle **310**. The second joint **622** is disposed to be vertically spaced apart from the first joint **621** along a height direction of the vehicle. The second joint **622** functions as a lower reference point of the kingpin shaft, which becomes a central axis on which the wheel **2** rotates during a steering operation of the wheel **2**. Accordingly, when the first knuckle **310** is rotated by the steering force transmitted from the first joint **621**, the second joint **622** may induce so that the lower side of the first knuckle **310** is rotated while maintaining the kingpin shaft.

(68) FIG. **8** is an enlarged view schematically illustrating a configuration of a second joint according to first embodiment of the present disclosure.

(69) Referring to FIGS. **1** to **8**, the second joint **622** according to the present disclosure may be connected to a lower end of the second knuckle **320** and supported by being combined to an end of the extension arm **622a** extending toward the first knuckle **310**. The second joint **622** is vertically spaced apart from the first joint **621** along the height direction of the vehicle and is connected to the lower side of the first knuckle **310**. Like the first joint **621**, the second joint **622** may be exemplified as various types of constant velocity joints.

(70) The first joint **621** and the second joint **622** are disposed to be inclined at a predetermined angle with respect to the ground so that the kingpin shaft may achieve a kingpin inclination angle of a certain size.

(71) FIG. **9** is a perspective view schematically illustrating a configuration of a corner module apparatus for a vehicle according to a second embodiment of the present disclosure, FIG. **10** is a front view schematically illustrating a configuration of the corner module apparatus for a vehicle according to the second embodiment of the present disclosure, FIG. **11** is an exploded perspective view schematically illustrating a configuration of the corner module apparatus for a vehicle according to the second embodiment of the present disclosure, and FIG. **12** is an exploded perspective view illustrating an angle adjustment unit in the corner module apparatus for a vehicle according to the second embodiment of the present disclosure.

(72) Referring to FIGS. **9** to **12**, a corner module apparatus **1** for a vehicle according to the present embodiment is configured to include a driving unit **100**, a first knuckle **310**, a second knuckle **320**, a suspension arm **410**, a joint arm **700**, and an angle adjustment unit **800**, which are described in detail as follows.

(73) The driving unit **100** is installed inside the wheel **2** of the vehicle, and rotates the wheel **2** by providing a driving force to the wheel **2**. The driving unit **100** is installed in each of the wheels **2** of the vehicle to provide driving force to a plurality of wheels **2** individually. The driving unit **100** according to the present embodiment may include a stator, fixed inside the wheel **2**, which forms a magnetic field by receiving power from a vehicle battery, a rotor, installed rotatably inside the wheel **2**, which rotates the wheel **2** by electromagnetic interaction with the stator. The stator and the rotor may be disposed such that the central axis thereof is positioned on the same line as the central axis of the wheel **2**, and may be concentrically stacked on top of each other on the inside of the wheel **2**.

(74) The first knuckle **310** is coupled to the driving unit **100**, transfers the steering force to the

wheel **2**, and rotates about the second knuckle **320** described below. More specifically, the first knuckle **310** provides a mechanical connection to the driving unit **100** and functions as a constituent element that constitutes a kingpin axis A, the central axis of steering, during an operation of steering the wheel **2**.

(75) The first knuckle **310** may be rotated by receiving the steering force generated from a steering actuator (not illustrated) or a steering wheel (not illustrated) from a tie rod (not illustrated). The first knuckle **310** according to the present embodiment may be coupled to and supported by a stator of the driving unit **100** using a bolt or the like. The first knuckle **310** may rotatably support the rotor of the driving unit **100** with a wheel bearing or the like in between. The first knuckle **310** may be manufactured by casting a metal-based material in a mold or the like to ensure sufficient rigidity.

(76) Each of both upper and lower ends of the first knuckle **310** are rotatably supported by the second knuckle **320** described below. Both ends of the first knuckle **310** are disposed to be inclined by a predetermined angle with respect to a Z-axis perpendicular to the ground. In this case, an angle by which both ends of the first knuckle **310** are inclined is set to be the same as an inclination angle of the kingpin axis A, the central axis for steering.

(77) A specific shape of the first knuckle **310** is not limited to the shape illustrated in FIGS. **9** and **10**, and various design changes may be made within the technical idea of a shape that may be coupled to the driving unit **100** and disposed to face an inner surface of the wheel **2**.

(78) The second knuckle **320** is disposed to face the first knuckle **310** and rotatably supports the first knuckle **310**. The second knuckle **320** functions as a constituent element that constitutes a suspension axis B that guides a bump and rebound behavior of the wheel **2** by providing a mechanical connection to the wheel **2** of a suspension arm **410** described below.

(79) Herein, a suspension axis B may be exemplified as an axis disposed parallel to a Z axis in FIGS. **9** and **10**.

(80) The second knuckle **320** is disposed to face the first knuckle **310** in the width direction of the vehicle. An upper end portion of the second knuckle **320** is connected to an upper end portion of the first knuckle **310** with a ball joint in between, and a lower end portion of the second knuckle **320** is connected to a lower end portion of the first knuckle **310** with a joint arm **700** described below in between.

(81) Accordingly, the suspension axis B formed in the second knuckle **320** and the kingpin axis A formed in the first knuckle **310** are separated from each other in the width direction of the vehicle, so that the kingpin axis A may be placed closer to the wheel **2**. Thus, a kingpin offset value may be reduced, and the driving and braking stability of the vehicle may be improved.

(82) A second fastening hole **325** corresponding to a first fastening hole **721** formed in a joint arm **700** described below is formed in the second knuckle **320** in a manner that passes through an outer surface of the lower end portion of the second knuckle **320**. A second slot hole **326** corresponding to a first slot hole **722** formed in the joint arm **700** in a manner that passes through the outer surface of the second knuckle **320**. In this case, the second slot hole **326** is formed in the shape of a longitudinal hole along the width direction of the vehicle.

(83) A guide unit **830** for guiding a washer **812** is formed on the outer surface of the second knuckle **320** so that a cam bolt **810** of the angle adjustment unit **800** described below rotates eccentrically. The guide unit **830** is formed in the shape of a concavely recessing a portion of arm A in which the second fastening hole **325** and the second slot hole **326** are formed. That is, the second fastening hole **325** and the second slot hole **326** are formed inside the guide unit **830** in a shape that surrounds the circumference of the second fastening hole **325** and the second slot hole **326**.

(84) The specific shape of the second knuckle **320** is not limited to the shape illustrated in FIGS. **9** and **10**, and various design changes may be made within the technical idea of a shape disposed to face the first knuckle **310** and rotatably support the first knuckle **310**.

(85) The suspension arm **410** extends from a vehicle body and absorbs shock or vibration applied to the wheel **2** from a road surface. More specifically, the suspension arm **410** supports the second

knuckle **320** with respect to the vehicle body, and at the same time, absorbs the load applied from the wheel **2** due to the rigidity of the suspension arm **410** during vehicle driving and controls movement of the wheel **2** when the wheel **2** bumps and rebounds.

(86) The suspension arm **410** is installed between the second knuckle **320** and the vehicle body. One end of the suspension arm **410** is rotatably connected to the vehicle body, and the other end thereof is disposed to face the second knuckle **320**. Herein, the vehicle body may be exemplified as a chassis frame such as a subframe (not illustrated) installed on a lower portion of the vehicle.

(87) The other end portion of the suspension arm **410** rotatably supports the second knuckle **320** with a separate connection member (not illustrated) in between. The suspension arms **410** are provided in a pair and are disposed to face each other while being spaced away from each other in the vertical direction.

(88) The other end portions of the pair of suspension arms **410** are connected to the upper and lower end portions of the second knuckle **320**, respectively. The pair of suspension arms **410** may be formed to have the shape of a double wishbone. Accordingly, the suspension arm **410** may set a negative camber of the wheel **2**, thereby improving cornering performance of the vehicle and performing low-floor setting to lower a vehicle height.

(89) A shock absorber **420** is provided in an expandable and contractable manner along the lengthwise direction thereof to absorb shock or vibration transferred from the road surface to the vehicle body through the wheel **2**.

(90) The joint arm **700** connects the lower end portion of the first knuckle **310** and the lower end portion of the second knuckle **320** to each other. The joint arm **700** may include a ball joint **710**, a fastener **720**, and a connection member **730**. The connection member **730** connects the ball joint **710** and the fastener **720** to each other.

(91) The ball joint **710** is joint-coupled to the lower end portion of the first knuckle **310** so that the first knuckle **310** may be rotatable.

(92) The fastener **720** is fastened to the second knuckle **320** in a state of being accommodated inward from the lower end portion of the second knuckle **320**.

(93) The first fastening hole **721** is formed in an upper portion of the fastener **720** in a manner that passes through an outer surface of the fastener **720**. The first slot hole **722** is formed in a lower portion of the fastener **720** in a manner that passes through the outer surface of the fastener **720**. In this case, the first slot hole **722** is formed in the shape of a longitudinal hole along the width direction of the vehicle.

(94) The angle adjustment unit **800** is provided between the second knuckle **320** and the joint arm **700**. The angle adjustment unit **800** adjusts a relative angle between the first knuckle **310** and the second knuckle **320** by axially rotating the joint arm **700** in the width direction of the vehicle.

(95) The angle adjustment unit **800** may be configured to include the cam bolt **810** and an adjustment member **820**.

(96) The washer **812** is eccentrically coupled to an outer circumferential surface of a bolt shaft **811** of the cam bolt **810**. The bolt shaft **811** passes through the first slot hole **722** and the second slot hole **326** in the cam bolt **810**.

(97) The adjustment member **820** is a bolt. The adjustment member **820** passes through the first fastening hole **721** and the second fastening hole **325** and fastens the second knuckle **320** and the fastener **720** to each other.

(98) A process of operating the corner module apparatus for a vehicle according to the present embodiment, configured as described, will be described below.

(99) FIG. **13** is a view illustrating a state where the corner module apparatus for a vehicle according to the second embodiment of the present disclosure is operated for zero camber setting. Referring to FIGS. **9**, **10**, **12**, and **13**, when the fastener **720** of the joint arm **700** is accommodated inward from the lower end portion of the second knuckle **320**, the adjustment member **820** passes through the first fastening hole **721** and the second fastening hole **325**, and a nut is spirally coupled to a free

end portion of the adjustment member **820**.

(100) The cam bolt **810** in which the washer **812** is eccentrically coupled to the bolt shaft **811** passes through the first slot hole **722** and the second slot hole **326**, and the nut is spirally coupled to a free end portion of the cam bolt **810**. In this case, a predetermined relative angle is maintained between the kingpin axis A and the suspension axis B.

(101) FIG. **14** is a view illustrating a state where the corner module apparatus for a vehicle according to the second embodiment of the present disclosure is operated for negative camber setting. Referring to FIGS. **9**, **10**, **12**, and **14**, when the cam bolt **810** is rotated in one (counterclockwise) direction while a rotating tool is coupled to a head of the cam bolt **810**, the cam bolt **810** is eccentrically rotated by the guide unit **830** surrounding the outer circumference of the washer **812** eccentrically coupled to the bolt shaft **811**. At the same time, the joint arm **700** is rotated about the adjustment member **820** from inward to outward in the width direction of the vehicle.

(102) That is, in a state where the suspension axis B is fixed, the inclination angle of the kingpin axis A is adjusted, and thus, a camber angle may also be adjusted. Accordingly, wheel alignments of the kingpin axis A and the suspension axis B may be individually adjusted.

(103) FIG. **15** is a view illustrating a state where the corner module apparatus for a vehicle according to the second embodiment of the present disclosure is operated for positive camber setting. Referring to FIGS. **9**, **10**, **12**, and **15**, when the cam bolt **810** is rotated in the opposite (clockwise) direction while the rotating tool is coupled to the head portion of the cam bolt **810**, the cam bolt **810** is eccentrically rotated by the guide unit **830** surrounding the outer circumference of the washer **812** eccentrically coupled to the bolt shaft **811**. At the same time, the joint arm **700** is rotated about the adjustment member **820** from outward to inward in the width direction of the vehicle.

(104) That is, in a state where the suspension axis B is fixed, the inclination angle of the kingpin axis A is adjusted, and thus, the camber angle may also be adjusted. Accordingly, the wheel alignments of the kingpin axis A and the suspension axis B may be individually adjusted.

(105) The corner module apparatus for a vehicle according to an embodiment of the present disclosure may adjust the inclination angle of the kingpin axis A by axially rotating the joint arm **700** connecting the lower end portion of the first knuckle **310** and the lower end portion of the second knuckle **320** to each other. Thus, the wheel alignments of the kingpin axis A and the suspension axis B may be individually adjusted.

(106) Various embodiments of the present disclosure do not list all available combinations but are for describing a representative aspect of the present disclosure, and descriptions of various embodiments may be applied independently or may be applied through a combination of two or more.

(107) A number of embodiments have been described above. Nevertheless, it will be understood that various modifications may be made. For example, suitable results may be achieved if the described techniques are performed in a different order and/or if components in a described system, architecture, device, or circuit are combined in a different manner and/or replaced or supplemented by other components or their equivalents. Accordingly, other implementations are within the scope of the following claims.

(108) While this disclosure includes specific examples, it will be apparent after an understanding of the disclosure of this application that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or

supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

Claims

1. A corner module apparatus for a vehicle, comprising: a driving unit installed inside a wheel, the driving unit being configured to provide a driving force to the wheel; a first knuckle coupled to the driving unit; a second knuckle spaced apart from the first knuckle in a width direction of the vehicle and disposed to face each other; a suspension unit configured to connect to the second knuckle and to support the second knuckle for a vehicle body; a steering driving unit installed in the second knuckle configured to generate a steering force; a steering angle adjustment unit connected to the first knuckle, the steering angle adjustment unit being configured to adjust a steering angle of the wheel in conjunction with the steering force generated from the steering driving unit, wherein the second knuckle comprises: a knuckle body disposed to face the first knuckle and provided with an accommodation unit into which the steering driving unit is inserted; a mounting unit which extends from a first side of the knuckle body, the mounting unit being configured to support the steering angle adjustment unit; a first connection unit extending from the mounting unit and connected to the suspension unit; and a second connection unit extending from a second side of the knuckle body and connected to the suspension unit, and wherein the first connection unit and the second connection unit are disposed to be spaced apart from each other in a direction perpendicular to a ground.
2. The corner module apparatus of claim 1, wherein the knuckle body is disposed to be inclined with respect to a ground.
3. The corner module apparatus of claim 1, wherein the accommodation unit penetrates the knuckle body in a width direction of the vehicle.
4. The corner module apparatus of claim 1, wherein the suspension unit comprises: a suspension arm provided between the second knuckle and the vehicle body and being configured to support the second knuckle; and a shock absorber connected to the suspension arm, the shock absorber configured to absorb an impact transmitted from a road surface.
5. The corner module apparatus of claim 4, wherein the suspension arm comprises: a first arm comprising a first end rotatably connected to the first connection unit and a second end rotatably connected to the vehicle body; and a second arm, the second arm being spaced apart from the first arm, the second arm comprising: a third end rotatably connected to the second connection unit; and a fourth rotatably connected to the vehicle body.
6. A corner module apparatus for a vehicle, comprising: a driving unit installed inside a wheel, the driving unit being configured to provide a driving force to the wheel; a first knuckle coupled to the driving unit; a second knuckle spaced apart from the first knuckle in a width direction of the vehicle and disposed to face each other; a suspension unit configured to connect to the second knuckle and to support the second knuckle for a vehicle body; a steering driving unit installed in the second knuckle configured to generate a steering force; and a steering angle adjustment unit connected to the first knuckle, the steering angle adjustment unit being configured to adjust a steering angle of the wheel in conjunction with the steering force generated from the steering driving unit, wherein the steering angle adjustment unit comprises: a deceleration unit configured to connect with the second knuckle and to slow down the steering force generated from the steering driving unit; and a joint unit configured to vary the steering angle of the wheel by transmitting the steering force from the deceleration unit to the first knuckle.
7. The corner module apparatus of claim 6, wherein the deceleration unit comprises: a first transmission gear configured to rotate with an input shaft of the steering driving unit; a second

- transmission gear that is configured to engage with the first transmission gear and to rotate in conjunction with a rotation of the first transmission gear; and a third transmission gear configured to engage with the second transmission gear and to rotate an output shaft in conjunction with a rotation of the second transmission gear.
8. The corner module apparatus of claim 7, wherein the first transmission gear comprises: a shape of a worm shaft; and worm threads provided on an outer peripheral surface of the first transmission gear.
9. The corner module apparatus of claim 7, wherein the joint unit comprises: a first joint extending from the output shaft and connected to a first side of the first knuckle; and a second joint extending from the second knuckle, spaced apart from the first joint, and connected to a second side of the first knuckle.
10. The corner module apparatus of claim 9, wherein the first joint and the second joint are disposed to be inclined at a predetermined angle with respect to a ground.
11. The corner module apparatus of claim 9, wherein the first joint and the second joint comprise constant velocity joints.
12. A corner module apparatus for a vehicle, comprising: a driving unit installed inside a wheel, the driving unit being configured to provide a driving force to the wheel; a first knuckle coupled to the driving unit; a second knuckle spaced away from the first knuckle to face the first knuckle in a width direction of the vehicle and being configured to rotatably support the first knuckle; a suspension arm configured to connect the second knuckle and a vehicle body and configured to support the second knuckle with respect to the vehicle body; a joint arm configured to connect the first knuckle and the second knuckle to each other; an angle adjustment unit provided between the second knuckle and the joint arm and configured to adjust a relative angle between the first knuckle and the second knuckle by axially rotating the joint arm in the width direction of the vehicle; a ball joint rotatably jointed to the first knuckle; a fastener fastened to the second knuckle; and a connection member configured to connect, the connection member being provided between the ball joint and the fastener.
13. The corner module apparatus of claim 12, wherein a first fastening hole is defined in the joint arm, the first fastening hole passing through an outer surface of the fastener, and wherein a first slot hole is defined in the joint arm, the first slot hole being positioned under the first fastening hole and passing through the outer surface of the fastener.
14. The corner module apparatus of claim 13, wherein a second fastening hole is defined in the second knuckle, the second fastening hole corresponding to the first fastening hole, and wherein a second slot hole is defined in the second knuckle, the second slot hole corresponding to the first slot hole.
15. The corner module apparatus of claim 14, wherein the first slot hole and the second slot hole are formed in a shape of a longitudinal hole in the width direction of the vehicle.
16. The corner module apparatus of claim 14, wherein the angle adjustment unit comprises: a cam bolt; a washer being eccentrically coupled to an outer circumferential surface of a bolt shaft of the cam bolt, wherein the first slot hole and the second slot hole pass through the bolt shaft; and a guide unit formed on an outer surface of the second knuckle, the guide unit being configured to guide the washer in such a manner as to eccentrically rotate the cam bolt.
17. The corner module apparatus of claim 16, wherein the angle adjustment unit further comprises an adjustment member passing through the first fastening hole and the second fastening hole, the adjustment member being configured to fasten the second knuckle and the fastener to each other.
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